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#task 1
import random

class Environment:
    def __init__(self):
        # Randomly assign dirt (0 = clean, 1 = dirty)
        self.locationCondition = {
            'A': random.randint(0,1),
            'B': random.randint(0,1),
            'C': random.randint(0,1),
            'D': random.randint(0,1)
        }

class SimpleReflexVacuumAgent:
    def __init__(self, environment):

        print("Initial Environment:", environment.locationCondition)

        score = 0

        # Random starting location
        vacuumLocation = random.choice(['A','B','C','D'])
        print("Vacuum starts at location:", vacuumLocation)

        locations = ['A','B','C','D']

        for loc in locations:

            if environment.locationCondition[loc] == 1:
                print("Location", loc, "is Dirty")
                environment.locationCondition[loc] = 0
                print("Cleaning Location", loc)
                score += 1
            else:
                print("Location", loc, "is already Clean")

        print("Final Environment:", environment.locationCondition)
        print("Performance Score:", score)

env = Environment()
agent = SimpleReflexVacuumAgent(env)
```

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Initial Environment: {'A': 1, 'B': 1, 'C': 0, 'D': 1}
Vacuum starts at location: D
Location A is Dirty
Cleaning Location A
Location B is Dirty
Cleaning Location B
Location C is already Clean
Location D is Dirty
Cleaning Location D
Final Environment: {'A': 0, 'B': 0, 'C': 0, 'D': 0}
Performance Score: 3
```

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#task 2
import random

class TrafficLightAgent:

    def __init__(self):

        lights = ["Red", "Yellow", "Green"]

        # Random light
        light = random.choice(lights)

        print("Traffic Light:", light)

        if light == "Red":
            print("Action: Stop")

        elif light == "Yellow":
            print("Action: Slow Down")
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elif light == "Green":  
    print("Action: Move")
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agent = TrafficLightAgent()
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Traffic Light: Yellow  
Action: Slow Down
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#task 3  
import random  
  
class AutomaticDoorAgent:  
  
    def __init__(self):  
  
        time = random.choice(["Day", "Night"])  
        personDetected = random.choice([True, False])  
        authorizedPerson = random.choice([True, False])  
  
        print("Time:", time)  
        print("Person detected:", personDetected)  
  
        if time == "Day":  
  
            if personDetected:  
                print("Door Action: OPEN")  
            else:  
                print("Door Action: CLOSE")  
  
        else: # Night security  
  
            print("Authorized person:", authorizedPerson)  
  
            if personDetected and authorizedPerson:  
                print("Door Action: OPEN")  
            else:  
                print("Door Action: CLOSED (Security Mode)")  
  
agent = AutomaticDoorAgent()
```

```
Time: Night  
Person detected: True  
Authorized person: True  
Door Action: OPEN
```